

Project Design Phase

Problem – Solution Fit Template

Date	18 June 2026
Team ID	
Project Name	ShopEz (Full Stack MERN Application)
Student Name	ALLADI MANJUNADH (alladimanjunadh07@gmail.com)
Maximum Marks	2 Marks

Problem – Solution Fit Framework:
Evaluating the alignment between a proposed technical architecture and verified user challenges is crucial. This framework ensures that ShopEz leverages targeted features to mitigate common online retail friction points, creating a tailored user experience.

- Purpose:**
- ☐ Solve complex digital purchasing obstacles for diverse user personas.
 - ☐ Maximize performance using the high-velocity MERN ecosystem.
 - ☐ Synchronize software development goals with user-centric requirements.
 - ☐ Build consumer trust through reliable data security and order transparency.
 - ☐ Modernize the e-commerce journey for the current digital era.

1. TARGET AUDIENCECS • Busy corporate professionals • Tech-savvy students • Families shopping for kids' wear • Small-scale retail administrators	6. USER CONSTRAINTSCC • Limited time for checkout • Unstable mobile internet • Desire for simple, clean UI • Concerns regarding payment safety	5. CURRENT ALTERNATIVESAS • Major global e-commerce portals • Direct sales on social media • Local brick-and-mortar shops • Legacy web shopping tools
2. CORE OBJECTIVESJ&P • Fast navigation to products • Consistent cart & order handling • Effective size/type filtering • Instant stock updates	9. ROOT ISSUESRC • Bloated and slow web stacks • Poor cart persistence • Cluttered navigation paths • Insecure data protocols	7. OBSERVED HABITSBE • Drops off if site is slow • Compares prices across tabs • Depends on refined search • Checks order status frequently
3. TRIGGERS & 4. STATESTR/EM • Reactive seasonal shopping • Fast load times boost intent • Motivated by relevant deals • Anxious about late deliveries	10. PROPOSED SOLUTIONSL • ShopEz: MERN-based SPA • High-speed React frontend • Admin tool for inventory control • JWT-based secure logins	8. ENGAGEMENT CHANNELSCH • Mobile-first React web app • Direct marketing campaigns • Protected backend portal • Automated mail/app alerts

References:

1. <https://www.ideahackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>